## 1. Explain the purpose of the Forget Gate in LSTM. Explain the formula:

Formula:
$f_t$= σ($W_f$ [$h_t$ −1, $x_t$] + $b_f$)
Explanation:
The Forget Gate decides how much information from the previous cell state should be kept or forgotten. It prevents the LSTM from storing unnecessary or outdated information.

Formula Explanation:
• $h_t$ −1 = Previous hidden state
• $x_t$= Current input
• [$h_t$−1, $x_t$] = Combination of previous hidden state and current input
• $W_f$ = Weight matrix of the Forget Gate
•$b_f$= Bias of the Forget Gate
• σ (Sigmoid) = Produces values between 0 and 1
  - 0 → Completely forget
  - 1 → Completely keep

The output $f_t$tells the LSTM how much of the previous memory should remain.

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## 2. Explain the role of the Input Gate in LSTM. Explain the formulas:

Formula 1:
it = σ(Wi[$h_t$−1, $x_t$] + bi)
Formula 2:
$\tilde{C}_t$= tanh(Wc[$h_t$−1, $x_t$] + bc)
Explanation:
The Input Gate decides which new information should be written into the cell state.
It has two parts:
(a) Input Gate
The sigmoid function produces it, which determines how much new information should be stored.
• 0 → Do not store
• 1 → Store completely
(b) Candidate Memory
The tanh function creates the candidate memory ($\tilde{C}_t$), which contains the new information learned from the current input.

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## 3. Explain how the cell state is updated in LSTM.

Formula:
Ct = $f_t$× Ct−1 + it × $\tilde{C}_t$

Explanation:
The cell state is updated in two steps.
Step 1:
The previous memory is multiplied by the Forget Gate.
$f_t$× Ct−1
This removes unnecessary information.
Step 2:
The Input Gate decides how much of the candidate memory should be added.
it × $\tilde{C}_t$

Finally:
• Old useful memory is kept.
• New important memory is added.
The updated memory becomes Ct.
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## 4. Explain the purpose of the Output Gate. Explain the formulas.

Formula 1:
$o_t$ = σ(Wo[$h_t$−1, $x_t$] + bo)
Formula 2:
$h_t$= $o_t$× tanh(Ct)

Explanation:
The Output Gate decides what information from the cell state should become the hidden state (output).
Formula 1:
The sigmoid function generates $o_t$, which decides how much information should be exposed.
Formula 2:
The updated cell state passes through tanh and is multiplied by the Output Gate.

This produces $h_t$, which is:
• The current output.
• The hidden state passed to the next time step.

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## 5. Why are three gates used in LSTM? Explain their responsibilities with formulas.

LSTM uses three gates because memory management requires three different decisions.
Forget Gate

Formula:
$f_t$= σ($W_f$ [$h_t$−1, $x_t$] + $b_f$)

Responsibility:
• Removes unnecessary old information.
Input Gate
Formulas:
it = σ(Wi[$h_t$−1, $x_t$] + bi)
$\tilde{C}_t$= tanh(Wc[$h_t$−1, $x_t$] + bc)

Responsibility:
• Selects useful new information.
• Stores it into memory.
Output Gate
Formulas:
$o_t$= σ(Wo[$h_t$−1, $x_t$] + bo)
$h_t$= $o_t$× tanh(Ct)

Responsibility:
• Selects useful information from memory.
• Produces the hidden state.

Together, these three gates enable the LSTM to forget irrelevant information, remember important information, and output only what is needed.

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## 6. Differentiate between Forget Gate, Input Gate, and Output Gate.

Forget Gate

Purpose:
• Removes unnecessary old memory.
Formula:
$f_t$= σ($W_f$[$h_t$−1, $x_t$] + $b_f$)

Input Gate
Purpose:
• Stores useful new information.
Formulas:
it = σ(Wi[$h_t$−1, $x_t$] + bi)
$\tilde{C}_t$= tanh(Wc[$h_t$−1, $x_t$] + bc)

Output Gate
Purpose:
• Produces the hidden state.
Formulas:
$o_t$= σ(Wo[$h_t$−1, $x_t$] + bo)
$h_t$= $o_t$× tanh(Ct)

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## 7. Explain how Forget Gate controls old memory and Input Gate controls new memory.

The Forget Gate determines how much of the previous cell state should be kept.

Formula:
$f_t$= σ($W_f$[$h_t$−1, $x_t$] + $b_f$)

It multiplies the previous memory:
$f_t$× Ct−1

to remove unnecessary information.
The Input Gate determines how much new information should be stored.
Formula:
it = σ(Wi[$h_t$−1, $x_t$] + bi)
The candidate memory is:
$\tilde{C}_t$= tanh(Wc[$h_t$−1, $x_t$] + bc)

Together:

it × $\tilde{C}_t$

adds the important new memory.
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## 8. Explain how the Output Gate determines the hidden state.

The Output Gate first computes:

$o_t$= σ(Wo[$h_t$−1, $x_t$] + bo)

which decides how much information from the updated memory should be revealed.
The updated memory passes through tanh.

Finally:
$h_t$= $o_t$× tanh(Ct)

produces the hidden state.
The hidden state is used for prediction and is passed to the next time step.

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## 9. "Forget Gate removes unnecessary information, Input Gate adds important information, and Output Gate controls the final output." Explain this statement with formulas.

Forget Gate
Formula:
$f_t$= σ($W_f$[$h_t$−1, $x_t$] + $b_f$)

Old memory retained:
$f_t$× Ct−1
Input Gate
Formulas:
it = σ(Wi[$h_t$−1, $x_t$] + bi)
$\tilde{C}_t$= tanh(Wc[$h_t$−1, $x_t$] + bc)

New memory added:
it × $\tilde{C}_t$

Cell State Update
$Ct = f_t \times Ct-1 + it \times \tilde{C}_t$

Output Gate
Formulas:
$o_t = \sigma(Wo[h_t-1, x_t] + bo)$
$h_t = o_t \times \tanh(Ct)$

Thus, the Forget Gate filters old memory, the Input Gate stores useful new memory, and the Output Gate determines the information sent as the hidden state.

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## 10. Explain the complete workflow of an LSTM cell using the formulas.

Step 1: Forget old information
$f_t = \sigma(W_f[h_t-1, x_t] + b_f)$

The Forget Gate decides what information from the previous cell state should be kept or discarded.

Step 2: Decide what new information to store
$it = \sigma(Wi\ [h_t-1, x_t] + bi)$
The Input Gate determines how much new information should be stored.

Step 3: Create candidate memory
$\tilde{C}_t = \tanh(Wc[h_t-1, x_t] + bc)$
A candidate memory containing new information is generated.

Step 4: Update the cell state
$Ct = f_t \times Ct-1 + it \times \tilde{C}_t$

The old memory is filtered, and the selected new information is added to form the updated cell state.

Step 5: Compute the Output Gate
$o_t = \sigma(Wo[h_t-1, x_t] + bo)$
The Output Gate decides how much of the updated memory should be revealed.

Step 6: Generate the hidden state

$h_t = o_t \times \tanh(Ct)$

The hidden state is produced and used as the output for the current time step and passed to the next LSTM cell.

Overall Workflow:

Input ($x_t$, $h_t$−1)
↓
Forget Gate
↓
Input Gate
↓
Candidate Memory
↓
Cell State Update
↓
Output Gate
↓
Hidden State ($h_t$)